

Constant-temperature anemometry in hypersonic flow: critical issues and sample results

E. F. Spina, C. B. McGinley

Abstract The critical issues are examined in the application of constant-temperature hot-wire anemometry to hypersonic boundary layers. While continuous turbulence measurements are more challenging to make in hypersonic flows, it is shown here that the difficulties can be overcome for a wide range of flow conditions. An extensive review of the literature reveals that many of the heat transfer complexities associated with hypersonic anemometry have already been resolved. Frequency-response tests, calibration results and boundary-layer measurements in hypersonic flow are also presented. A hot-wire frequency response of about 500 kHz was obtained in a Mach 11 flow, and the resulting boundary-layer spectra are smooth and repeatable.

List of symbols

D	hot-wire diameter
E	instantaneous anemometer voltage
f, g	functions of the overheat ratio, τ
f_r	hot-wire frequency response
$g(s)$	dimensionless function of the molecular speed ratio
k	thermal conductivity coefficient
l	active length of the hot-wire element
Kn	Knudsen number ($Kn \equiv \lambda/D$)
n	exponent of the $Nu-Re$ relation for a hot wire
Nu	Nusselt number
Pr	Prandtl number

R	electrical resistance
R_1	resistance of the anemometer bridge leg in series with the wire
Re	Reynolds number based on the hot-wire diameter
s	molecular speed ratio ($u/V_{mp} = \sqrt{\frac{\gamma}{2}} M$)
t	shock thickness
T_e	wire recovery temperature
U	freestream velocity
V_{mp}	most-probable molecular speed
α	thermal accommodation coefficient
δ	boundary layer thickness or shock detachment distance
η	recovery factor ($\eta \equiv T_e/T_0$) or Kolmogorov length scale
λ	mean free path
ρ	density
τ	hot-wire overheat ratio ($\tau \equiv (T_w - T_e)/T_0$)

Subscripts

e	boundary-layer edge value
w	wire value
∞	freestream condition
o	stagnation condition

1

Introduction

This paper provides a review of the critical issues in the application of constant-temperature hot-wire anemometry to hypersonic flows, with an emphasis on heat transfer and frequency response issues. Discussions of probe construction and hot-wire calibration techniques are included, and preliminary results from Mach 6 and Mach 11 flows are presented. The results include hot-wire calibrations, freestream spectra, and spectra and r.m.s. profiles across a Mach 11 boundary layer.

While laser Doppler velocimetry (LDV) and particle image velocimetry (PIV) are rapidly becoming the standard for turbulence measurements, thermal anemometry has some advantages over such particle-scattering techniques, particularly in high-speed flow. Both LDV and PIV suffer from seeding bias, wherein the non-negligible mass of the seed particles biases the velocity distribution toward a value that is lower than the actual distribution. Furthermore, the seeds may not accurately follow the small scales of motion in supersonic and hypersonic flows. If the objective is to obtain *continuous* data in high-speed flows,

Received: 30 November 1993/Accepted: 10 May 1994

E. F. Spina
Syracuse Center for Hypersonics,
Syracuse University,
Syracuse, NY 13244, USA

C. B. McGinley
Fluid Mechanics and Acoustics Division
NASA-Langley Research Center
Hampton, VA 23681, USA

Correspondence to: E. F. Spina

The work of the first author was supported by NASA-Langley Grant NAG-1-1400, under the supervision of Dr. Stephen K. Robinson. The assistance of Mr. Mark Sheplak in acquiring some of the data presented in § is gratefully acknowledged.

A version of this paper was an invited presentation at the *Third International Symposium on Thermal Anemometry*, sponsored by ASME, June 1993, Washington, D. C.

the limited data rate of LDV is a critical weakness. While an essentially continuous record can be acquired via LDV in low-speed flows, the seed density required for "instantaneous" data in a supersonic flow is excessive. At best, then, LDV is a tool for measuring mean turbulence quantities in supersonic flow. While PIV can provide velocity information in a planar field, it is limited to two-dimensional flows, and suffers from data-rate limitation and limited dynamic range. While a thorough investigation of a hypersonic boundary layer may eventually include a role for both of these techniques, the main objective of this study is an evaluation of constant-temperature anemometry (CTA) in such flows.

The pioneering research on supersonic turbulence in the 1950's was performed exclusively with constant-current anemometers (CCA) because of the simplicity of the electronic circuit and the ability to achieve a high frequency response (see, for example, Kovaszny 1950, 1954, Morkovin 1956, Laufer and McLellan 1956, and Kistler 1959). The frequency response of the CCA system is maintained with an R-C compensation unit, as compared to the feedback control system needed for constant-temperature operation. The high-bandwidth, high-gain, low-noise amplifiers needed to produce stable constant-temperature circuits have been available only for the past twenty-five years. Automatic compensation for the thermal inertia of the wire is the main advantage of the CTA system over constant-current operation, where the compensation must be set manually.

There are applications where constant-current operation is still preferred over the constant-temperature mode, however. The anemometer output in high-speed flows is sensitive both to stagnation-temperature fluctuations, T'_0 , and to mass-fluctuations, $(\rho u)'$. At wire temperatures scarcely above T_0 (low overheat ratio, $\tau \equiv (T_w - T_0)/T_0$), a hot-wire anemometer is primarily sensitive to T'_0 , while it is primarily sensitive to $(\rho u)'$ at higher overheat ratios. Anemometer response to T'_0 is highly non-linear at low overheat, and the very large gain needed in the CTA feedback loop can create stability problems. Furthermore, the frequency response of CTA systems degrades significantly as the overheat ratio is decreased. The constant-temperature system is thus a poor choice to make T'_0 measurements, and CCA systems are preferred in such applications. On the other hand, when an anemometer is operated at high overheat ratio, $\tau \approx 1$, the contribution of T'_0 to the anemometer output can be neglected (Kovaszny 1950 and Smits et al. 1983). If the measurement of mass-flux fluctuations is the primary goal of the experiment, constant-temperature operation is preferred because of the automatic compensation of the hot wire. Furthermore, operation at high overheat precludes the need to make measurements at numerous overheat ratios, and continuous, instantaneous mass-flux can be measured.

The application of thermal anemometry to supersonic flow poses a number of operational difficulties, however. For example, the high dynamic pressures and high stagnation temperatures common in hypersonic laboratory flows have a direct effect on wire lifetime. High-temperature flows (such as the Mach 6 air flow discussed in §3) mandate the use of wire materials that have a higher oxidation temperature than that of tungsten, but are also weaker. For example, Pt-Rh(10%) has an oxidation temperature that is almost twice as high as that of tungsten, but it has a yield strength that is only 10% that of

tungsten. The shocks emanating from the wire supports must also be avoided, necessitating the use of plated hot-wire elements. The high frequency content of the turbulent motions in supersonic and hypersonic flow puts extraordinary demands on the anemometer circuitry and increases the possibility of unstable or unbalanced systems. Foremost among the difficulties associated with supersonic anemometry, however, is that heat transfer is being measured and velocity only inferred through a correlation that is particularly complex for compressible flows. The details of the heat-transfer correlation will be discussed at length in the following section.

2

Critical issues in hypersonic anemometry

Since thermal anemometry techniques only provide for direct measurement of heat transfer from a fine heated wire, a theoretical or empirical relation is needed to determine the flow velocity from the measured quantity. The general functional statement of the heat-transfer relation for a cylinder in high-speed crossflow was given by Kovaszny (1950):

$$Nu = Nu\left(\frac{l}{D}, M, Re, Pr, \tau, \eta\right), \quad (1)$$

The three parameters Re , Kn , and M are related by $Kn = \sqrt{\frac{\pi}{2}} M_\infty / Re$, so only two are needed to describe the heat-transfer relation in high-speed flow. The dependence on wire aspect ratio is important only at low values of l/D . At the aspect ratios commonly used (200 or greater), end-conduction effects are small and are accounted for in a mass-flux calibration (although Perry et al. 1979 show that end-conduction effects may be important for the dynamic response). In most flows where anemometry can be applied, the Prandtl number will be constant.

If radiation and end-conduction effects are negligible or can be corrected for, then the heat transfer from a hot wire that is one leg of a balanced anemometer bridge can be described by the semi-empirical relation (Smits et al. 1983):

$$Nu = \frac{E^2 R_w}{\pi k l (R_1 + R_w)^2 (T_w - \eta T_0)} = Af(\tau) + Bg(\tau) Re^n, \quad (2)$$

where A and B are constants for a given wire, independent of the operating point. In low-speed flow this relation is termed King's Law, and $n = 0.5$; ρ , k , and μ are all constant; $\eta = 1$; and $f = g = 1$, implying a linear dependence on wire temperature. The $Nu-Re$ relation is not as simple in hypersonic flow. The Reynolds-number exponent and recovery factor are both functions of Knudsen number, the fluid properties and density are no longer constant, there are several temperatures representative of the flowfield, and the dependence on overheat ratio is more complex. Many of these complexities were examined in a series of papers in the 1950's and 1960's, and those results will be reviewed and summarized here as a preliminary to presentation of the recent hot-wire calibration results.

2.1

The flowfield around a hot-wire at hypersonic speeds

Following Kovaszny (1950), Fig. 1 shows schematics of the flow around 5 μm hot wires at two different supersonic Mach numbers. The sketches provide a physical sense of the significant

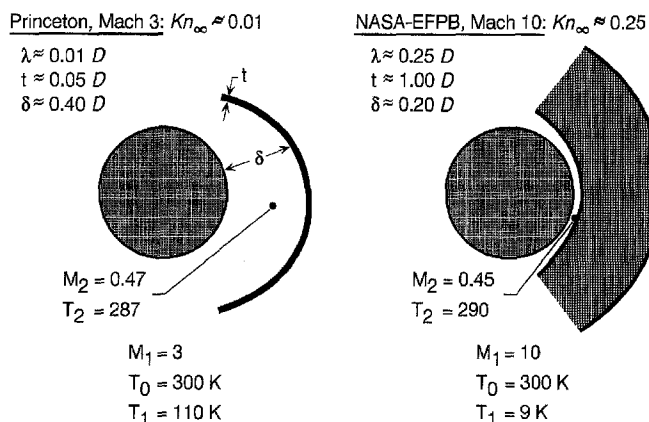


Fig. 1. Approximate schematics of the flowfield around $5\ \mu\text{m}$ hot wire elements in two supersonic flows: a A Mach 3 air flow with $T_0 = 300\ \text{K}$ (Princeton University), b A Mach 10 helium flow with $T_0 = 300\ \text{K}$ (NASA-Langley)

differences between moderately supersonic and hypersonic flows around a cylinder. A $5\ \mu\text{m}$ -diameter cylinder in a Mach 3 air flow with a unit Reynolds number of $6.3 \times 10^7\ \text{m}^{-1}$ has an upstream Knudsen number¹ (λ/D) of 0.01, placing it near the limit of continuum flow and into the slip flow regime. The shock thickness for the Mach 3 case is 3–5 mean-free paths, or about 5% of the wire diameter in this flow (taking shock thickness as the distance over which translational equilibrium is achieved). Since the shock detachment distance is about 35% of the wire diameter (Liepmann and Roshko 1957), there is a distinct region between the detached shock and the cylinder. This is in dramatic contrast to a Mach 10 helium flow with a unit Reynolds number of $2 \times 10^7\ \text{m}^{-1}$ around the same hot wire. The Knudsen number for this situation is about 0.2, placing it well within the slip-flow regime and on the verge of free-molecular flow. As a result, the shock wave thickness is about $1.0D$ and the change in properties can no longer be considered discontinuous. The larger shock thickness and a smaller shock detachment, $0.2D$, blur the distinction between properties upstream of the wire and downstream of the shock wave.

Although the temperature varies considerably in the flow around the hot wire, analysis requires that a single temperature be chosen at which to evaluate flow parameters and fluid properties. Some rationale can be provided for using any one of three temperatures: the upstream temperature characteristic of the flowfield, T_∞ , the value downstream of the bow shock wave and in the vicinity of the hot wire, T_2 , and the (constant) stagnation temperature, T_0 . Of course, for supersonic flows T_0 and T_2 are nearly equal: T_0/T_2 decreases to 1.05 by Mach 3 in air and asymptotes toward 1.03 for $M_\infty \rightarrow \infty$.

Kovaszny (1950) found that, for $M_\infty < 2$, the Nusselt number could be correlated as a unique function of the Reynolds number when all flow parameters are evaluated behind the normal shock. However, μ and k vary slowly with temperature (in air, $\mu \sim k \sim T^{0.768}$) and little difference is observed if T_0 is used instead of T_2 . Stalder et al. (1951b) also showed that the Nusselt number is a function of the Reynolds number alone if μ and

k are evaluated at T_0 for a wide range of flow conditions ($0.025 \leq Kn_\infty \leq 11.8$). Similar results were indicated by Baldwin et al. (1960) in correlating the majority of heat-transfer data from transverse cylinders taken in high-speed flows prior to 1960 (Mach numbers to 6). As for the physical rationale for using T_0 , Kovaszny offered that "since the heat transfer takes place mainly on the blunt part of the cylinder where almost stagnation conditions prevail, it is reasonable to evaluate heat conduction and viscosity under stagnation conditions." Dewey (1961) noted that at low Reynolds number, a distinct shock wave does not appear in front of the hot wire (witness the Mach 10 schematic). It is therefore unlikely that the temperature downstream of a normal shock wave can describe the region in front of the cylinder. Finally, since stagnation temperature can be measured independently, it is most convenient to use T_0 in both calibrating and analyzing hot-wire data. Unless noted otherwise, then, the viscosity and thermal conductivity will be evaluated at the stagnation temperature and the Reynolds and Nusselt numbers will be denoted Re_0 and Nu_0 , respectively.

2.2

Mach-number independence of the Nu - Re relation

While the general functional statement of the heat-transfer relation for a cylinder in high-speed crossflow (Eq. 1) indicates a Mach-number dependence, the widely used semi-empirical relation (Eq. 2) does not. This is somewhat curious, since some Mach-number effects are intuitively expected in compressible flows.

Baldwin et al. (1960) correlated much of the cylinder heat-transfer data and found that significant Mach-number effects occur from $M = 0.2$ – 0.9 , with $Nu \sim \sqrt{1 + 0.2M^2}$. They also plotted all the supersonic cylinder heat-transfer data available in 1960 and found a clear Mach number independence over the range $1.3 \leq M_\infty \leq 6.1$. Their plot is reproduced here as Fig. 2, showing data from nine different investigators with a scatter of less than 15%. This summary plot includes the data of Laufer and McClellan (1956), who found no variation in the Nu - Re relation for the Mach number range they tested, $1.3 \leq M_\infty \leq 4.5$. The independence of the heat-transfer relation (and also the recovery factor) from Mach number is due to the hypersonic freeze phenomenon or the 'Mach-number independence principle' (see, for example, Hayes and Probstein 1966). Both the viscous and inviscid flowfields become independent of M as $M_\infty \rightarrow \infty$. The onset of Mach-number independence below the relatively low Mach number of 2 is due to the bluntness of the body. The stronger normal shock created by the blunt two-dimensional cylinder will determine the flowfield characteristics in the stagnation region, which in turn is where the majority of the wire heat transfer occurs. Simply, for a given upstream Reynolds number, there is very little difference in stagnation conditions as the shock strength increases above that at Mach 2.

2.3

Reynolds-number exponent

The exponent of the Nu - Re correlation (Eq. 2) is of great practical importance when reducing hot-wire anemometer data. Both theory and experiment have been used in attempts to determine a functional relationship between Re (or Kn) and n . While no unifying theory exists, there is good agreement

¹ Following convention, the Knudsen number is formed using properties upstream of the hot wire and its bow shock unless noted otherwise.

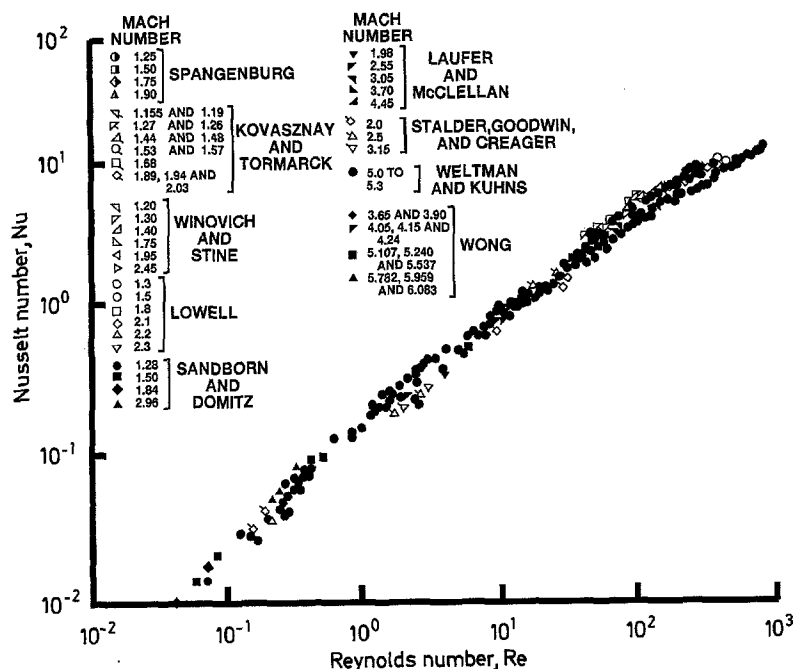


Fig. 2. Summary of supersonic heat transfer from transverse cylinders (Baldwin et al. 1960). ©1960 ASME – reprinted with permission

between data sets and theory for low and high Knudsen number limits.

When the flow Knudsen number is very high, say on the order of 10, the heat transfer can be described in terms of molecular motions only. The flow in this free-molecular regime can be analyzed by applying corrections to continuum flow equations or by solving the Boltzmann equation for the molecular velocity distribution (i.e., finding corrections to the Maxwell distribution). Both methods are perturbation techniques, with the solutions in terms of the Knudsen number and the reciprocal of the Knudsen number, respectively. The perturbation methods fail for $Kn_\infty \sim O(1)$, however, and empirical results must be relied upon exclusively in this slip flow regime.

Stalder et al. (1951b) used kinetic theory to develop the relation for heat transfer from a cylinder in free-molecular flow:

$$Nu_0 = C_1 \alpha Re_0 Pr \frac{g(s)}{s},$$

where $s (\equiv u/V_{mp} = \sqrt{\frac{1}{2}} M)$ is the molecular speed ratio and $g(s)$ is a dimensionless function of s . The Prandtl number is constant over a wide temperature range for gases, and $g(s)/s$ is nearly constant for $s > 2$ (and therefore for M greater than about 2.4 in air). Thus,

$$Nu_0 \propto Re_0,$$

and the Reynolds-number exponent in free-molecular flow is predicted to be 1.0.

When the Knudsen number is low (high Re and moderate M), boundary-layer theory can be used to predict the flowfield near the stagnation point of the cylinder. Cole and Roshko (1954) used the Oseen approximation to develop an expression for the Nusselt number in subsonic continuum flow, and found a Reynolds-number exponent of 0.5. Beckwith (1959) showed that the exponent was 0.5 in high-Reynolds-number supersonic flow as well. Many data sets are in excellent agreement with these approximate solutions, and while they are theoretically

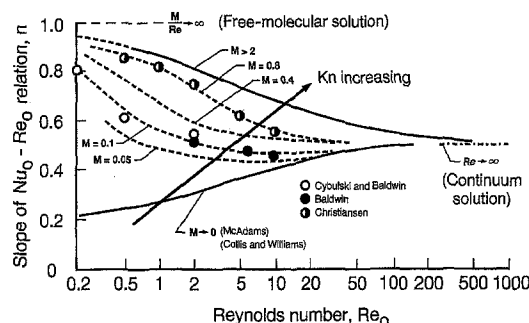


Fig. 3. Slope of the Nusselt-Reynolds number correlation showing the effect of Knudsen number (Dewey 1961). ©1961 AIAA – reprinted with permission

valid for $Re_0 \gg 1$, some researchers (Laufer and McClellan, for example) have found that the continuum results extend as low as $Re_0 \approx 20$. Dewey (1961) indicates that the departure from the continuum value of the exponent does not occur until $Re_0 = 200$, however.

Figure 3 (from Dewey) shows the free-molecular solution ($Nu \sim Re^1$), the continuum solution ($Nu \sim Re^{0.5}$), and data from the slip flow regime (where $Re \sim O(1)$ and $M \sim O(1)$). As the Knudsen number goes from continuum values to the slip flow range, the Reynolds-number exponent deviates from 0.5 and increases with increasing Knudsen number until free-molecular flow is reached.

2.4

Recovery factor (η)

The recovery temperature is the temperature that an unheated wire assumes when placed in a compressible flow, and it is needed to specify the $Nu-Re$ relation properly (see Eq. 2). The difference between this recovery temperature and the heated wire temperature provides the driving potential for convective heat transfer. In low-speed, continuum flow, the recovery temperature is equal to the stream stagnation temperature. In

high-speed, continuum flow, viscous dissipation causes gases (which have $Pr < 1$) to have a recovery temperature less than the stagnation temperature. Data from Kovaszny (1950), Laufer and McClellan (1956), Baldwin et al. (1960), and Dewey (1961) indicate that the recovery factor is independent of Mach number in supersonic flow as long as the flow remains in the continuum regime. The lack of functional dependence of η on the Mach number is another result of the Mach-number independence principle. In continuum flow, the measured values of η range from 0.95 to 0.97 for a wide range of supersonic Mach numbers.

The recovery temperature in free-molecular flow is, somewhat surprisingly, greater than the stagnation temperature of the flow. This is explained by Stalder et al. (1951a,b) through a consideration of the magnitudes of the incident and re-emitted molecular energy. They show that the energy of a free-molecular flow incident on a blunt body is the same as would be found in continuum flow for large values of the molecular-speed ratio (therefore large M and Kn). Furthermore, for a specified re-emitted stream temperature, less energy per molecule is transported from a body in free-molecular flow than in continuum flow. Thus, if the same total amount of energy (i.e., the incident energy) is to be removed in each case, the body temperature must be higher for a free-molecular flow than for a continuum flow. Theory developed by Stalder et al. (1951a,b) predicts that $\eta \rightarrow 2\gamma/(\gamma + 1)$ for high-Mach-number, free-molecular flow. The experimental data support Stalder's theory; an extensive correlation given by Baldwin et al. is provided here as Fig. 4. The figure illustrates that all of the continuum flow data has $\eta < 1$ (regardless of Mach number), and all of the free-molecular flow data has $\eta > 1$. The transition from data that follows the continuum flow trend to that which agrees with free-molecular theory occurs in the range $0.2 < Kn_\infty < 5$. Only in this range is the recovery factor functionally dependent upon Mach number.

2.5

Dependence of anemometer response on overheat ratio

Since any variation in stagnation temperature is reflected in the value of the hot-wire overheat ratio ($\tau = \tau(T_0, \eta)$), the dependence of the wire response on overheat ratio must be found for applications where the stagnation temperature is not constant. Most experiments in high-speed flow fall into this category, including cases where the hot-wire calibration and the data acquisition are not done at the same stagnation temperature, or when the mean stagnation temperature varies during a data run.

In subsonic flow, the anemometer response exhibits a linear dependence on the temperature loading, $T_w - T_e$. Since $Nu \sim (T_w - T_e)^{-1}$, this results in a constant slope and intercept for King's Law (corresponding to $f(\tau) = g(\tau) = 1$ in Eq. 2) for a given wire at a particular overheat ratio. While several researchers have observed a non-linear anemometer response to overheat ratio in supersonic flow, only Kovaszny (1950) and Smits et al. (1983) have quantified the relationship. Kovaszny was the first to note the non-linearity in supersonic flow for large temperature loadings, and he found the functional form:

$$Nu \sim (T_w - T_e)(1 - C\tau),$$

where C is a constant (equal to 0.18) found by calibrating the hot wire at various overheat ratios. Kovaszny hypothesized that the larger temperature loading influences the detachment distance of the shock wave, thereby affecting property changes across the shock.

Smits et al. (1983) defined the functional dependence of the Nusselt number on overheat through the functions f and g which appear in Eq. 2. They found $f(\tau) = 1 - 0.65\tau$ and $g(\tau) = 1 - 0.085\tau$. In light of the fact that shock position is related to the overheat dependence of the anemometer, it is not surprising that Smits et al. and Kovaszny measured different

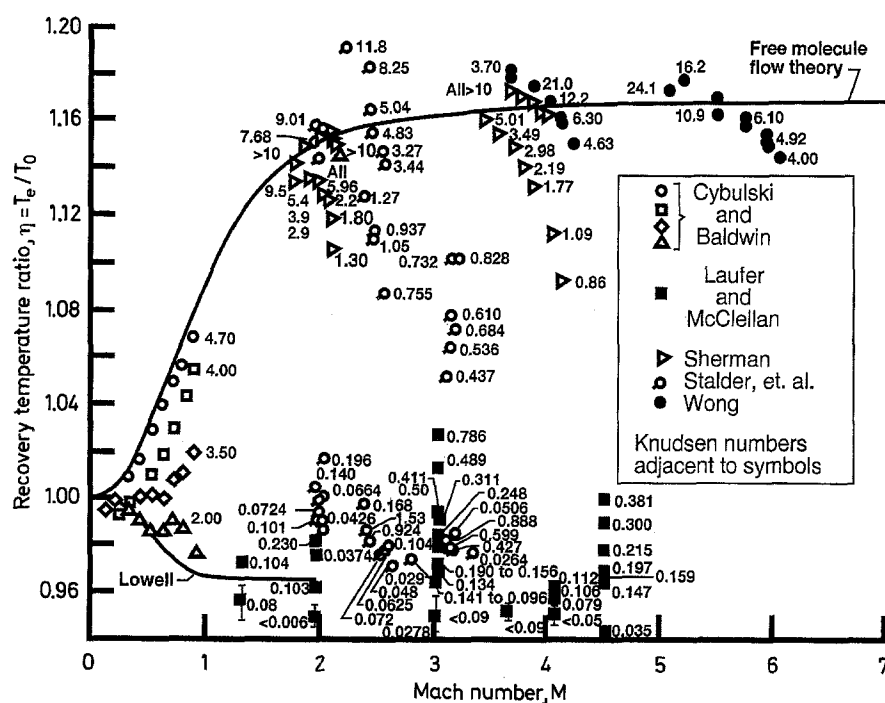


Fig. 4. Recovery factor as a function of Mach number for a wide range of Knudsen numbers. Note that the Knudsen number for each data point is adjacent to the symbol (Baldwin et al. 1960). ©1960 ASME – reprinted with permission

correction factors. Any differences in the probe geometry will affect the shock position, including the amount of slack in the wire, the length of the active element, and the geometry of the wire supports. Thus, a certain variation in overheat dependence is to be expected since the design of supersonic hot-wire probes differs greatly from one research lab to another.

2.6

Frequency response and spatial resolution requirements

The dynamic response of a hot wire in supersonic flow was reviewed by Smits (1990). The demands on the frequency response and spatial resolution of an anemometer system in hypersonic flow are even more severe than discussed in Smits' review because of the higher Mach number and Knudsen number. The bottom line is, of course, how much of the turbulence spectrum is resolved by the hot wire. Based on measurements in supersonic flow, Kistler (1959) suggested that the frequency response must exceed $5U_e/\delta$ to measure $\sqrt{u'^2}$ within 10% in the outer part of the boundary layer. Because of the scarcity of hypersonic turbulence data and uncertainty about Mach-number and Reynolds-number effects, the only way to be certain that the hot-wire frequency response is high enough is to check the measured spectrum. If the roll-off of the spectrum occurs near the frequency response of the wire, then the validity of the measurements may be questionable and the frequency response should be increased.

High-Reynolds-number flows can have a Kolmogorov scale that is much smaller than the active length of the hot wire ($\eta \ll l$), and the high-frequency end of the spectrum will therefore be distorted. The classic study of hot-wire spatial resolution was conducted by Wyngaard (1968) for subsonic flow, but his results are at least approximately applicable to supersonic flow based upon Morkovin's hypothesis (1962) and the work of Spina et al. (1991). Wyngaard showed that when $\eta/l \rightarrow 0$, the spectrum measured by the hot wire falls to one-half its true value at a wave number of $2.1/l$. It is important to evaluate the value of $2.1/l$ for the flow to be measured so that this limit on the validity of the data is clearly identified.

3

Results from hypersonic flow

The experimental study was performed in hypersonic wind tunnels at NASA-Langley Research Center (within the Experimental Flow Physics Branch). Two different series of tests were conducted, one in a Mach 10/11 helium flow, the other in a Mach 6 air flow. The Mach 11 boundary layer measurements were made on a 4° wedge in the 1.5 m diameter Mach 20 High-Reynolds-Number Helium Tunnel. A description of the Mach 20 tunnel is given in Watson and Bushnell (1971), and details on the mean flow over the wedge can be found in Watson (1978). The tunnel was operated at a stagnation pressure of $57 \times 10^5 \text{ N/m}^2$ (800 psi) and ambient stagnation temperature, resulting in a unit Reynolds number of $19.7 \times 10^6 \text{ m}^{-1}$. The 4° wedge is 2.29 m long, and the measurements were taken at a point on the centerline 2.16 m downstream of the leading edge, where $\delta \approx 51 \text{ mm}$, $Re_\theta \approx 6800$, and $M_e \approx 11$. Run times for this facility are very short (about 5 seconds), so in-situ hot-wire calibration is not feasible. The hot-wires were instead calibrated in a 76 mm-diameter Mach 10 helium pilot tunnel. The mass-flux

sensitivity of the hot wires was determined by placing the wires in the freestream and varying the stagnation pressure from $6.9 \times 10^5 \text{ N/m}^2$ (100 psi) to $57 \times 10^5 \text{ N/m}^2$ (800 psi) with ambient T_0 . The run-time of the pilot tunnel was long enough so that the operating characteristics of the hot wires could be thoroughly investigated.

Because the mean properties of the Mach 6 boundary layer have not yet been characterized, only hot-wire calibration results will be presented for this flow. The full capabilities of the Mach 6 High-Reynolds-Number Tunnel are presented in Jones and Williams (1970). For these tests, the blowdown tunnel was operated at a nominal stagnation temperature of 505 K and stagnation pressures between $6.9 \times 10^5 \text{ N/m}^2$ (100 psi) and $34.4 \times 10^5 \text{ N/m}^2$ (500 psi); this resulted in unit Reynolds numbers ranging from $5.72 \times 10^6 \text{ m}^{-1}$ to $28.62 \times 10^6 \text{ m}^{-1}$. Under these conditions, run-times were sufficiently long to allow for in-situ hot-wire calibration. A 0.91 m-long flat plate was placed on the centerline of the 0.3 m-diameter test section. Mass-flux calibrations of the hot wire were performed in the freestream above the plate by varying the stagnation pressure between $6.9 \times 10^5 \text{ N/m}^2$ and either $20.7 \times 10^5 \text{ N/m}^2$ or $34.4 \times 10^5 \text{ N/m}^2$. While an attempt was made to hold the stagnation temperature constant, it typically increased by about 2% as p_0 passed from $6.9 \times 10^5 \text{ N/m}^2$ to $10.35 \times 10^5 \text{ N/m}^2$.

3.1

Hot-wire frequency response

Due to the need for a high frequency response in these flows, great care was taken in the design and construction of the hot-wire probes. The basic probe design follows that outlined by Fernando (1988). Of particular importance was the elimination of as many cable connectors as possible; the frequency response appears to degrade as more connectors are used (particularly the miniature coaxial connectors). Kovasznay's (1950) method of plating and soft-soldering the wire, and then etching an active portion, is favored to avoid shock interference from the wire supports. $5 \mu\text{m}$ -diameter wire was used for most of the measurements, with an active length ranging from 0.8 to 1.0 mm, so that $l/d \geq 160$. Tungsten wire was used in the helium flows, but the high flow stagnation temperature and the relatively low oxidation temperature of tungsten ($\sim 600\text{K}$) prevented its use in the Mach 6 tunnel. Platinum-Rhodium wire (a 10% alloy) was used in the hot-air flow instead, and the material properties resulted in much more frequent wire breakage and lower wire sensitivity.

The optimum frequency response was achieved in both flows using symmetric (1:1) anemometer bridges with an external arm. The DISA 55M12 and the TSI IFA 100 bridges provided system frequency responses that were superior to those achieved with standard 10:1 or 20:1 bridges. The frequency response of the hot wires was determined using the standard square-wave injection technique. The highest frequency response achieved in the Mach 10 flow was 700 kHz, with a typical value of about 500 kHz (see Fig. 5). The system frequency response was verified via direct laser-beam heating (similar to the method of Bonnet and de Roquefort 1980). These measurements agreed with the frequency response inferred from the square-wave test to within 5%. A frequency response of 500 kHz or more is excellent, particularly when compared to Kistler's criterion of $5U_e/\delta$, which is 200 kHz for the Mach 11 boundary layer. The

high-frequency electronic “strain gauging” that is so common with hot wires in high-speed flows was, in most cases, completely alleviated by mounting the wires with slack. Any data contaminated by strain gauging were discarded. As for the spatial resolution, according to Wyngaard’s criterion (1968), the measured one-dimensional spectrum will fall to one-half its true value at about 800 kHz in this flow. The true limitation on high-frequency spectral measurements is therefore the hot-wire frequency response, not the size of the active element.

The frequency response in the Mach 6 flow was lower than that observed at Mach 10, and the wires suffered from severe strain gauging when the frequency response was maximized. The maximum frequency response was found to be about 200 kHz, but the response had to be “detuned” to approximately 175 kHz to reduce the strain gauging. While strain gauging is usually attributed to fluctuations in the wire tension due to high-frequency variations in the aerodynamic loading, no amount of slack could eliminate the strain gauging in this case. As discussed by Smits and Dussauge (1989), strain gauging seems also to depend upon the electronic characteristics of the anemometer. The interaction between these two factors is highly complex, and it is difficult to make a determination as to why strain gauging affected the Mach 6 hot wires but not the Mach 11 wires.

3.2 Hot-wire calibration

The heat-transfer relation was explored in greater depth at Mach 10 than at Mach 6 because of the relatively low operating cost of the helium calibration tunnel and the high demand for Mach 6 tunnel time. The complete Mach 10 results will be discussed first, and then the $Nu-Re$ relation will be presented for the Mach 6 flow.

The recovery factor of the 5 μm wires at Mach 10 was determined by placing the unheated wires in the flow and measuring the resistance at various values of the stagnation pressure. Once the resistance was measured, the recovery

temperature was calculated from

$$R_e = R_a [1 + \alpha (T_e - T_a) + \beta (T_e - T_a)^2],$$

where α and β are the first and second temperature coefficients of resistivity, and T_a and R_a are the ambient temperature and resistance of the wire (see Morkovin 1956). Results for five different runs are shown in Fig. 6 for a Knudsen number range of 0.07 to 0.25, which is within the slip-flow regime. The recovery factor lies between 0.96 and 1.0 throughout the relatively small Knudsen range, although each run has nearly constant η . Although the correlation given by Baldwin et al. (Fig. 4) does not include data from Mach numbers as high as 10, the Mach-number independence principle allows a direct comparison with data at equivalent Knudsen number and any Mach number greater than 2. The current data compares favorably with that displayed by Baldwin et al. The cause of the run-to-run variation of η is unknown, but a $\pm 2\%$ variation from an average value of 0.98 will not affect the final results significantly.

The variation of the hot-wire Nusselt number with Reynolds number was investigated by operating the Mach 10 calibration tunnel over a range of stagnation pressures at fixed hot-wire overheat ratio. Reduced data from a representative hot-wire calibration is plotted in Fig. 7 (with a Reynolds-number

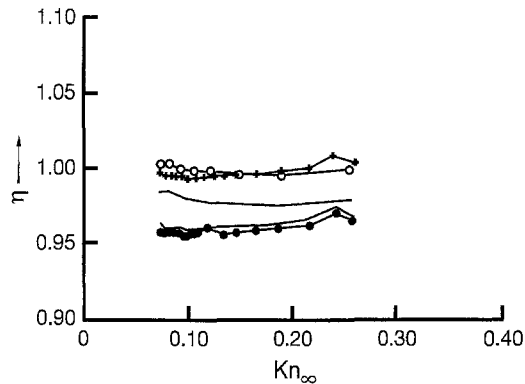


Fig. 6. Recovery factor, η , for a 5 μm wire in the Mach 10 calibration flow. Each symbol represents a different hot-wire element

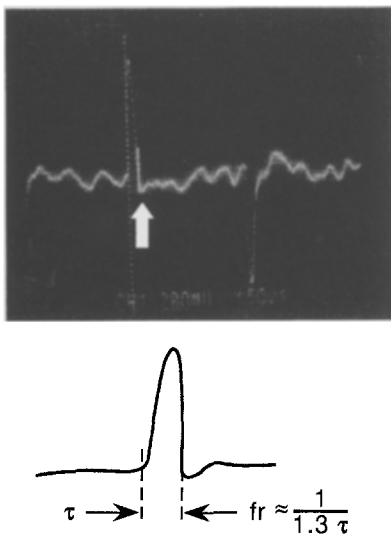


Fig. 5. Polaroid of the response of a tungsten hot wire to a square-wave perturbation in a Mach 10 flow. The time between the cursor is 1.5 μs , implying that the frequency response is approximately 500 kHz

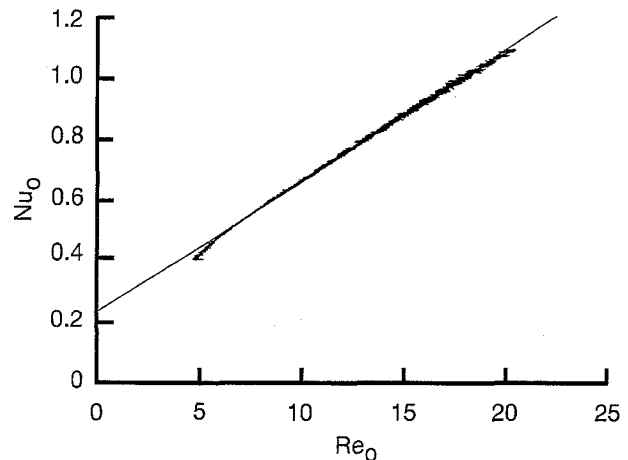


Fig. 7. Plot of the heat-transfer relation for a 5 μm wire in the Mach 10 calibration flow. The overheat ratio is about 0.9 for this run

exponent of 1.0). The data compares well to the summary plot given by Baldwin et al. and supplied here as Fig. 2. The small quantitative difference between their summary data and that shown in Fig. 7 is probably due to the variation caused by the overheat ratio (see below), as well as to the fact that no correction was made for end-conduction effects. The Reynolds-number exponent in Eq. 2, n , was determined for these flow conditions through a curve-fit routine. The best-fit value was 0.73, which lies along the curve for $M > 2$ and $5 \leq Re_0 \leq 20$ in Fig. 3. A practical hot-wire calibration is not expressed as Nu vs. Re^n , however, but as E^2 vs. $(\rho U)^n$, so that a measured anemometer voltage can be converted directly to a mass-flux value. An example of such a calibration curve for the Mach 10 flow is provided as Fig. 8, where $n = 0.73$.

As discussed in §2, the $Nu-Re$ relation exhibits a non-linear variation with temperature loading. The values of $f(\tau)$ and $g(\tau)$ in Eq. 2 were investigated for the Mach 10 flow by conducting wire calibrations at different values of the overheat ratio. There was a clear and consistent decrease in the slope of the $Nu-Re^n$ relation with increasing overheat ratio. It was calculated that $g(\tau) = 1 - 0.5\tau$ (Fig. 9), which is a more dramatic change than Kovaszny (1950) found, but not as sharp as Smits

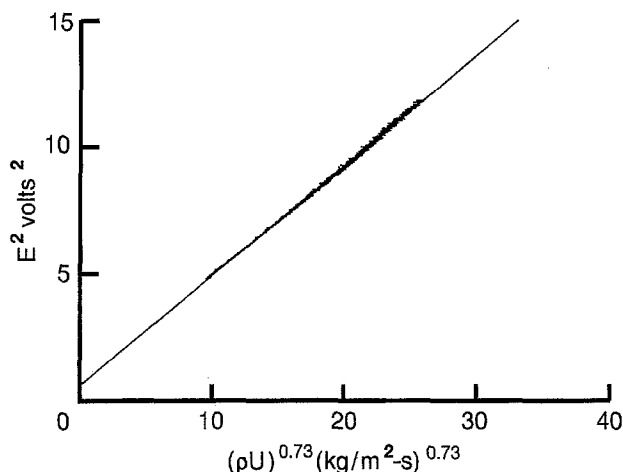


Fig. 8. Mass-flux hot-wire calibration from the Mach 10 helium tunnel

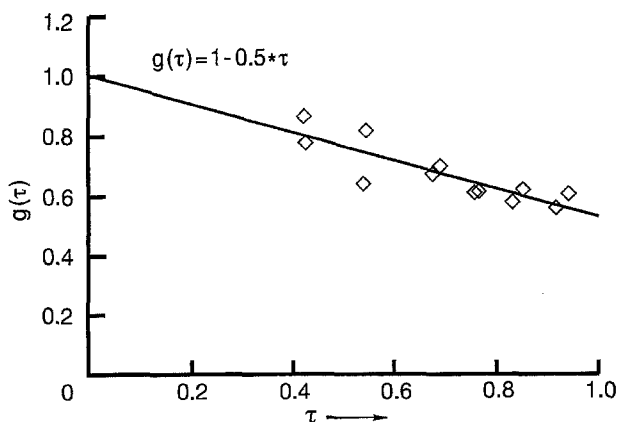


Fig. 9. The correction function for variation of anemometer response with overheat ratio

et al. (1983) measured. These data do not shed any light on the physics of this phenomenon, since there are no apparent Mach-number or Reynolds-number trends exhibited in a comparison of the data from Kovaszny, Smits et al., and the current study. As for the variation of the $Nu-Re^n$ intercept with overheat ratio, the current data are inconclusive. The Reynolds number range examined in this flow ($Re_0 = 5$ to 20) is considerably lower than that examined by Kovaszny ($Re_0 = 50$ to 250) and Smits et al. ($Re_0 = 80$ to 200). Because of this, the intercept value of the Nusselt number is so close to zero that the measurement uncertainty overwhelms the increase or decrease of the intercept with varying overheat ratio. To a first approximation, then, $f(\tau) = 1$.

The $Nu-Re$ relationship was also developed for the Mach 6 flow. The calibration was done by varying the tunnel stagnation pressure with the hot wires located in the freestream of the tunnel. The Reynolds-number exponent for the best linear fit to this data is about 0.58, which is slightly lower than expected for this Reynolds-number range (Fig. 3). While an attempt was made to hold the stagnation temperature constant, T_0 did increase slightly during the calibration. While T_0 variation is accounted for when calculating the Nusselt and Reynolds numbers, a T_0 correction such as shown by Smits et al. must be made to the data when the calibration is done in terms of the anemometer voltage and measured mass flux. The correction involves knowledge of the functions $f(\tau)$ and $g(\tau)$, which have not yet been determined for this flow.

3.3

Sample boundary layer measurements

The hot-wire anemometers were operated at an overheat ratio of approximately one for the Mach 11 boundary-layer investigation. As mentioned in §1 and as discussed at length by Kovaszny (1950), Smits et al. (1983), and Smits and Dussauge (1989), the effects of the stagnation temperature fluctuations on the hot wire may be neglected when $\tau \approx 1$. The hot-wire anemometer thus primarily responds to mass-flux fluctuations. The character of the mean mass-flux profile in a hypersonic boundary layer is dominated by the density variation across the layer. For the Mach 11 helium boundary layer, the density at the edge of the boundary layer is roughly 35 times greater than that at the wall. The large density gradients across hypersonic boundary layers lead to significant density fluctuations, which in turn dominate the mass-flux fluctuations sensed by the hot-wire anemometer.

After calibrating both 2.5 μm and 5.0 μm wires in the Mach 10 calibration tunnel, they were tested in the Mach 11 boundary layer. The 2.5 μm wires did not survive the starting shock, but the 5.0 μm wires proved to be extremely durable, never breaking at the test stagnation pressure ($55 \times 10^5 \text{ N/m}^2$). The power spectral density of the mass-flux fluctuations is provided in Fig. 10 for eight locations within the boundary layer. The system roll-off is about 350 kHz for this particular hot wire, and this appears to be more than adequate to capture the majority of the motions contributing to the r.m.s. of the fluctuating mass flux. The power spectral density at the location nearest the wall does exhibit considerable spikes around 100 kHz, but this is most probably electronic noise that overpowers the very small signal at the wall. Although not shown here, the run-to-run repeatability of the spectra was excellent across the entire dynamic range.

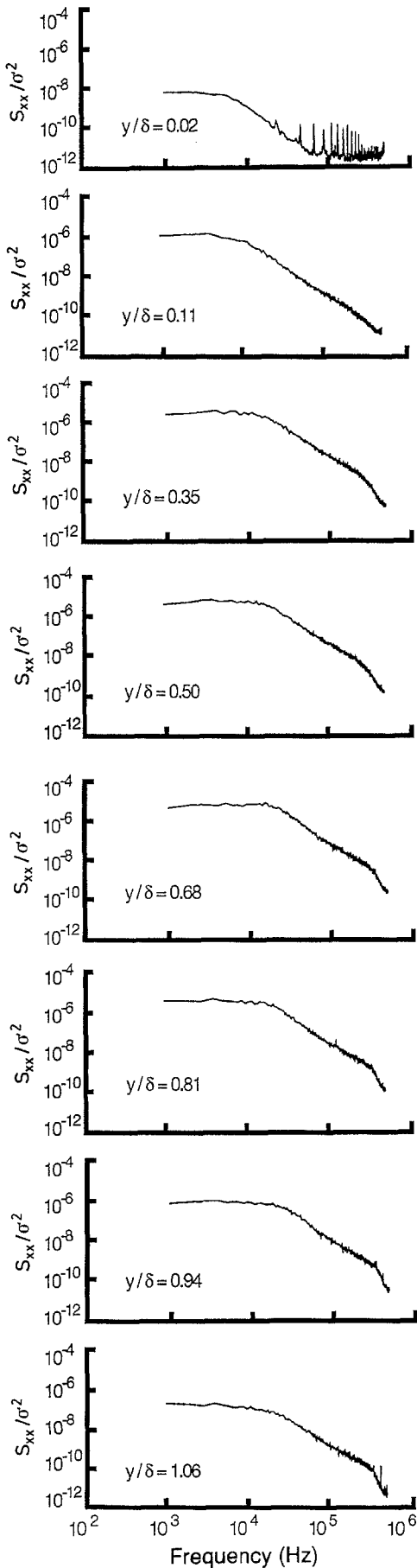


Fig. 10. Mach 11 helium boundary layer spectra at various positions in the flow

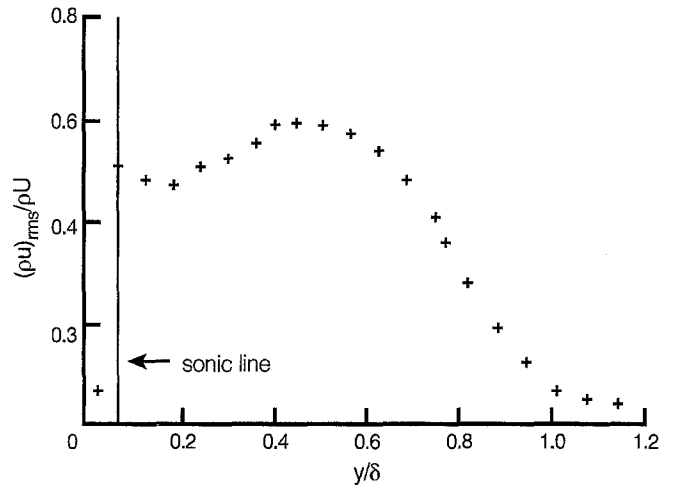


Fig. 11. Fluctuating mass-flux profile from the Mach 11 helium boundary layer

The corresponding distribution of the r.m.s. mass-flux fluctuations normalized by the local mean mass flux is presented in Fig. 11¹. The most remarkable features of this profile are the reduced level of fluctuations near the wall and the very large peak far from the wall (compared to lower-speed boundary layers). These features are typical of hypersonic boundary layers and are both due to the low density near the “hot” wall and the severe mean density gradient across the boundary layer.

A discrepancy exists between the mean mass flux measured by the hot-wire and that determined from pitot surveys; the difference reaches 20% near the middle of the boundary layer ($(\rho u)_{\text{pitot}} > (\rho u)_{\text{hw}}$). While the cause of this difference is as yet unknown, the very high level of fluctuations in this boundary layer may cause the overprediction of the mass flux as measured by a pitot-pressure probe. It does not appear that this discrepancy is caused by hot-wire calibration drift, since most hot wires are calibrated both before and after a series of runs in the boundary-layer tunnel. In any case, normalizing the r.m.s. fluctuations by the hot-wire mean mass flux will minimize the error from any such incorrect measurement. For a detailed discussion of other sources of uncertainty in normal-wire measurements in supersonic flow, the reader is referred to the discussion by Smits and Dussauge (1989).

4

Conclusions

The application of thermal anemometry to hypersonic boundary layers was reviewed and extended in this contribution. Conclusions will be offered on the current state of knowledge and on the promise for accurate, instantaneous measurements of mass flux in hypersonic boundary layers.

The review of heat-transfer issues in supersonic anemometry indicates that most relevant problems have been resolved, even in the slip-flow regime encountered in most hypersonic flows. The Mach-number independence of the flowfield around the blunt wire simplifies the heat-transfer relation significantly. For example, both the recovery factor and the $Nu-Re$ slope are

¹ The sonic line in this flow is at $y \approx 0.07\delta$; any data below this point is questionable due to the Mach-number sensitivity of hot wires there.

exclusive functions of the Knudsen number when the Mach number is greater than about 2. As a result, excellent empirical correlations exist for both of these important parameters, with well-defined asymptotes in the limits of continuum and free-molecular flow. While there was some confusion in the 1960's as to which temperature properly characterizes the heat transfer in the supersonic flow-field, the stagnation value is clearly preferred as the temperature at which to evaluate fluid properties. One issue that would benefit from further analysis is the effect of overheat ratio on the heat-transfer relation. Three researchers have now examined this effect, and no clear Mach-number, Knudsen-number, or Reynolds-number trend exists. Furthermore, there does not even appear to be a satisfactory flow model to explain the variation of the $Nu-Re$ correlation with temperature loading of the wire. The understanding of this issue and the entire heat-transfer problem would be enhanced by a CFD solution of the flowfield around a heated cylinder placed in a supersonic stream.

The present measurements have shown that constant-temperature anemometry can successfully be applied to hypersonic flows. Modern electronics have greatly improved the CTA frequency response, and the hot-wire response in the Mach 11 flow was considerably higher than the spectra roll-off frequency. The calibration of the hot wires for mass-flux sensitivity produced heat-transfer correlations in terms of the Nusselt and Reynolds number that agreed with previous results from constant-current anemometers. Measurements in the Mach 11 helium boundary layer yielded smooth, spike-free spectra and repeatable r.m.s. distributions. Future work in this experimental program will include measurements at higher Reynolds number (doubling p_0), measurements in the transitional region of the model, and further work in the Mach 6 air boundary layer.

References

- Baldwin LV; Sandborn VA; Laurence JC (1960) Heat transfer from transverse and yawed cylinders in continuum, slip, and free molecule air flows. *J Heat Trans* 82: 77–86
- Beckwith IE (1959) Similar solutions for the compressible boundary layer on a yawed cylinder with transpiration cooling. NASA TR R-42
- Bonnet JP; de Roquefort TA (1980) Determination and optimization of frequency response of constant temperature hot-wire anemometers in supersonic flows. *Rev Sci Instrum* 51: 234–239
- Cole J; Roshko A (1954) Heat transfer from wires at Reynolds numbers in the Oseen range. *Proc Heat Transfer Fluid Mech Inst, Univ. Cal. Press*, pp 13–23
- Dewey CF (1961) Hot wire measurements in low Reynolds number hypersonic flows. *J Amer Rocket Soc* 28: 1709–1718
- Fernando EM (1988) Supersonic turbulent boundary layer in an adverse pressure gradient. Ph.D. Thesis, Department of Mechanical and Aerospace Engineering, Princeton University
- Hayes WD; Probstein RF (1966) Hypersonic flow theory. Academic press.
- Jones RA; Williams VF (1970) Preliminary surveys of the wall boundary layer in a Mach 6 axisymmetric tunnel. NASA TN D-562
- Kistler AL (1959) Fluctuation measurements in a supersonic turbulent boundary layer. *Phys of Fluids* 2: 290–296
- Kovácsnay LSG (1950) The hot-wire anemometer in supersonic flow *J Aeronautical Sciences* 17: 565–573
- Kovácsnay LSG (1954) Development of turbulence-measuring equipment. NACA Report 1209
- Laufer J; McClellan R (1956) Measurements of heat transfer from fine wires in supersonic flows. *J Fluid Mech* 1: 276–289
- Liepmann HW; Roshko A (1957) Elements of gasdynamics, GALCIT Aeronautical Series, John Wiley & Sons
- Morkovin MV (1956) Fluctuations and hot-wire anemometry in compressible flows. AGARDograph 24
- Morkovin MV (1962) Effects of compressibility on turbulent flows. Intl Symposium on the Mechanics of Turbulence, A. Favre, Ed., CNRS, Paris, pp 367–380
- Perry AE; Smits AJ; Chong MS (1979) The effects of certain low frequency phenomena on the calibration of hot-wires *J Fluid Mech* 90: 415–431
- Smits AJ; Hayakawa K; Muck KC (1983) Constant-temperature hot wire anemometer practice in supersonic flows – Part 1: The Normal Wire. *Exp in Fluids* 1: 83–92
- Smits AJ; Dussauge J-P (1989) Hot-wire anemometry in supersonic flow. In AGARDograph AG-315, Chapter 5
- Smits AJ (1990) An introduction to hot-wire anemometry in supersonic flows. Symp. on the Heuristics of Thermal Anemometry, 1990 ASME FED Meeting, Toronto
- Spina EF; Donovan JF; Smits AJ (1991) On the structure of high-Reynolds-number supersonic turbulent boundary layers. *J Fluid Mech* 222: 293–327
- Stalder JR; Goodwin G; Creager MO (1951a) A comparison of theory and experiment for high-speed free-molecule flow. NACA Report 1032
- Stalder JR; Goodwin G; Creager MO (1951b) Heat transfer to bodies in a high-speed rarefied-gas stream. NACA TN 2438
- Watson RD; Bushnell DM (1971) Calibration of the Langley Mach 20 high Reynolds number helium tunnel including diffuser measurements. NASA TM X-2353
- Watson RD (1978) Characteristics of Mach 10 transitional and turbulent boundary layers. NASA TP-1243
- Wyngaard JC (1968) Measurements of small-scale turbulence structure with hot wires. *J Phys E: Sci Instr* 1: 1105